

Johnny Rodriguez

Data 622

Assignment 2

Model Performance Analysis for Client Subscription Prediction Using Decision Tree, Random Forest and AdaBoost Modeling

This report details the results from a series of experiments designed to identify an optimal classification model for predicting client subscription to a term deposit, based on a banking dataset. The experiments focused on three machine learning algorithms: Decision Trees, Random Forest, and AdaBoost. The experiments evaluated the impact of key hyperparameters on model performance, with a specific focus on managing the bias-variance tradeoff to achieve high predictive accuracy. The results indicate that an optimized Random Forest model provides the highest accuracy, while a constrained Decision Tree offers significant value for its interpretability.

Experiment Selection

To ensure direct comparison between models, we established a consistent experimental framework: all models were trained and evaluated using the exact same one-hot encoded dataset and the same static train-test split. This baseline environment eliminates data-centric variables, ensuring that observed differences in performance are attributable to the algorithms themselves and their hyperparameter configurations.

The experiments were then chosen to compare model behavior, progressing from a simple, interpretable model to more complex ensemble methods. This approach allows for analysis of how each model performs under comparable / similar conditions.

- **Decision Tree Experiments (Baseline and Pruning):** The initial experiments focused on the Decision Tree algorithm to establish a baseline and demonstrate the bias-variance tradeoff.
 - First, an unconstrained Decision Tree was trained. As expected, this model is prone to growing until it perfectly classifies the training data, leading to high variance and overfitting.
 - To counteract its tendency to overfit, two pruning techniques were tested. The first experiment constrained the tree by setting `max_depth=4`, directly limiting its complexity. The second experiment optimized for `min_samples_leaf`, finding an optimal value of 35. These methods intentionally simplify the model to reduce variance and improve its ability to generalize to unseen data.
- **Ensemble Method Experiments (Random Forest and AdaBoost):** The choice to tune the `n_estimators` hyperparameter for both Random Forest and AdaBoost was deliberate. It provides

a direct comparison of how the two different ensemble approaches, bagging and boosting, leverage an increasing number of base estimators.

- Random Forest uses bagging to reduce variance by averaging the predictions of many deep, uncorrelated decision trees. The experiment tuned `n_estimators` (the number of trees) to find the optimal point where predictive power stabilizes before the marginal gains diminish, finding an optimal value of 140 estimators. We are also able to hone-in and further hypothesize that a combination of controlled estimators and `max_depth` could further improve the model, as evidenced by the second experiment which found a depth of 25 improves Random Forest performance.
- AdaBoost uses boosting to build a strong classifier by sequentially training weak learners (shallow trees) and placing more weight on instances that were previously misclassified. Tuning its `n_estimators` (number of weak learners) finds the optimal balance before the model begins to overfit. The experiment identified the optimal number of learners as 120. Here too, we note that the learning rate also regularizes the model and we can hypothesize that a combination of learning rate and weaker learner tuning would further optimize the model.

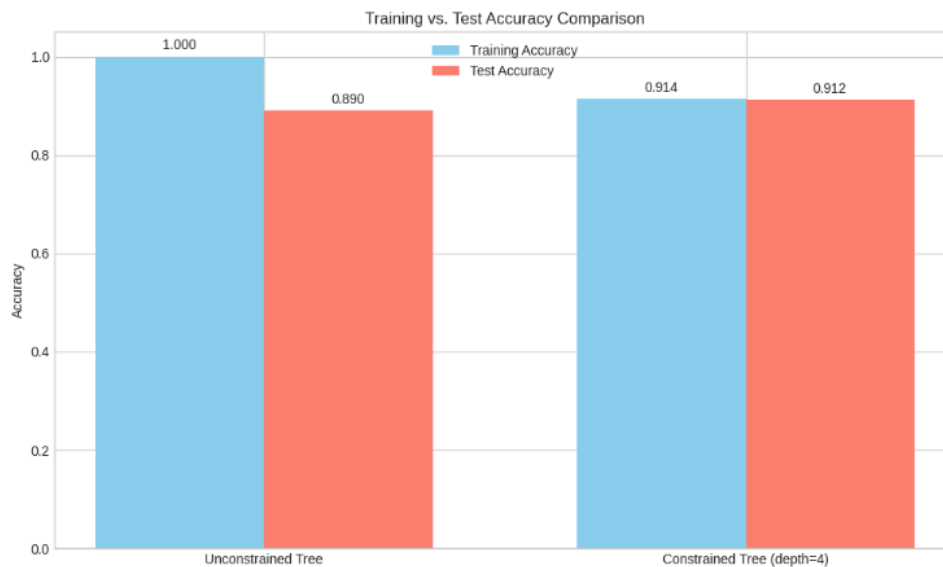
Analysis of Bias and Variance Across Models

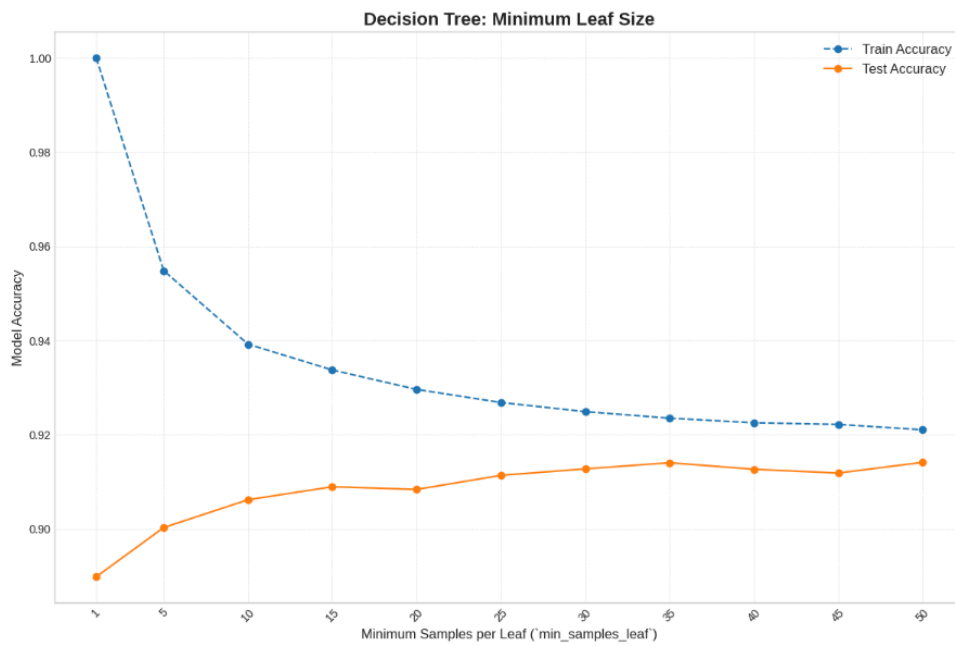
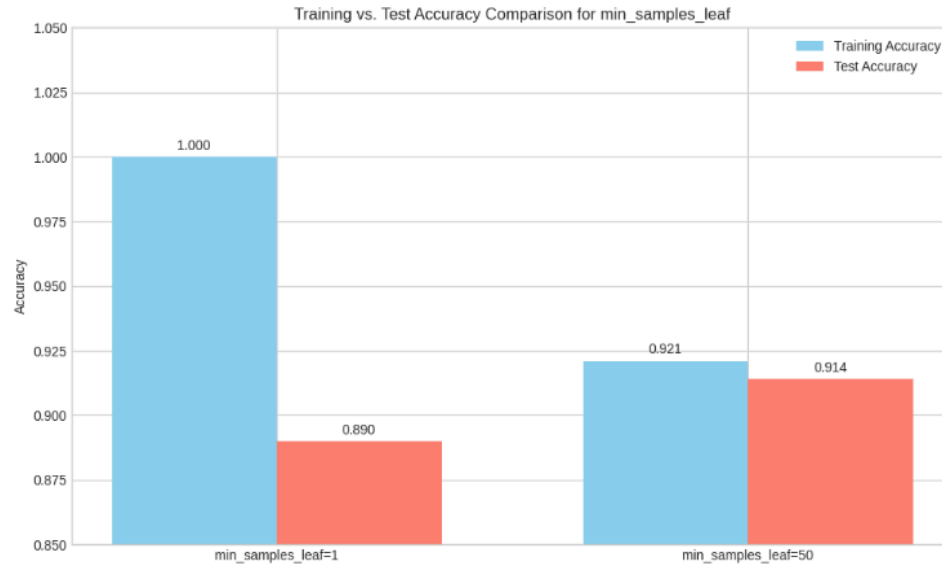
Bias-variance tradeoffs were a central consideration across the experiments.

Decision Trees: The first experiment contrasted a high-variance model with a lower-variance one. The unconstrained Decision Tree exhibited classic signs of overfitting; its perfect performance on the training data did not translate to the test set, resulting in an accuracy of only 0.8899. This indicates low bias but high variance. By constraining the tree depth to 4, we intentionally introduced a small amount of bias, making the model less complex. This simplification resulted in a significant reduction in variance, leading to a much-improved test accuracy of 0.9122. Similarly, increasing the minimum leaf size forced the model to generalize more, again reducing variance and improving performance.

Decision Tree Results

The Decision Tree visualizations demonstrate the bias-variance tradeoff. The unconstrained models (max_depth=None and min_samples_leaf=1) show perfect training accuracy (1.0) but significantly lower test accuracy, a sign of high variance and overfitting. By constraining the tree's depth to 4 or increasing the minimum leaf size, we introduce a small amount of bias, as the model no longer perfectly fits the training data. This trade-off dramatically reduces the variance, evidenced by the much smaller gap between the training and test scores. Ultimately, this leads to more reliable performance on unseen test data.

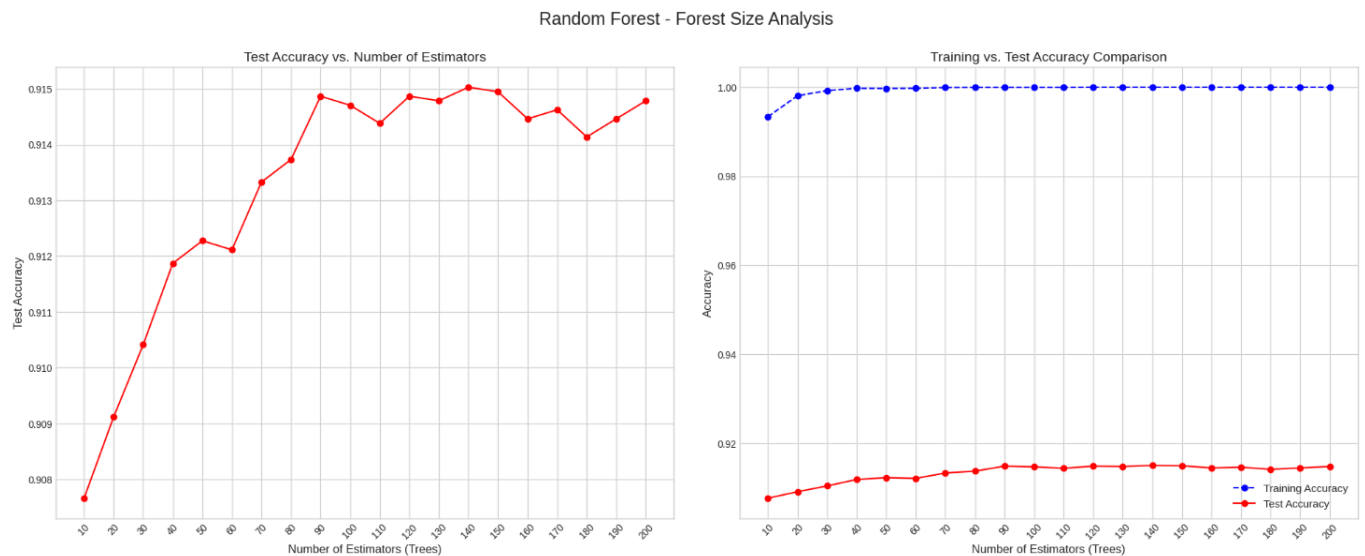




The Random Forest model is designed to tackle the high variance of individual decision trees. By building multiple trees on different subsets of data and features and then averaging their predictions, it produces a model with significantly lower variance than a single unconstrained tree. As seen in the results, the optimized Random Forest achieved the highest accuracy (0.9160), demonstrating its effectiveness in maintaining low bias while achieving low variance as we tested how many trees would be needed to optimize its performance.

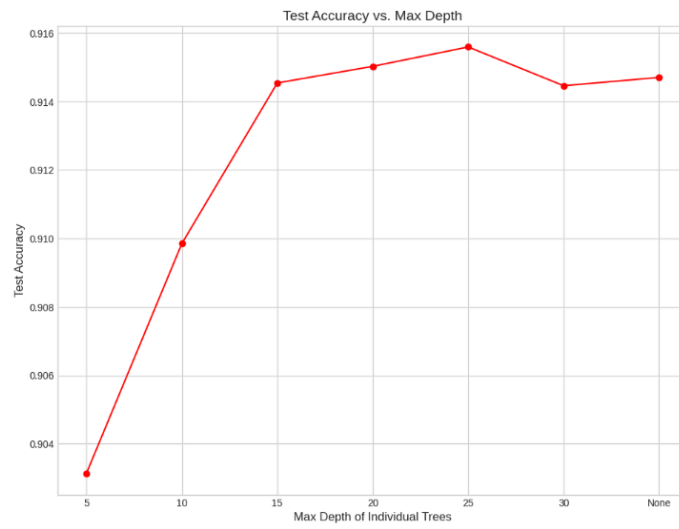
Random Forest Results

The Random Forest experiments demonstrate the balance between variance and bias through hyperparameter tuning. The first visualization shows that increasing the number of trees ($n_estimators$) rapidly improves test accuracy by averaging out individual errors, but performance plateaus around 100 estimators, revealing a point of diminishing, fluctuating, and marginal returns.



The second visualization illustrates that controlling the complexity of each tree (max_depth) to a depth of around 25 provides the best balance; shallower trees underfit (high bias) and unconstrained trees overfit (high variance), memorizing the training data but failing to generalize as effectively.

Random Forest Complexity Analysis (n_estimators=100)

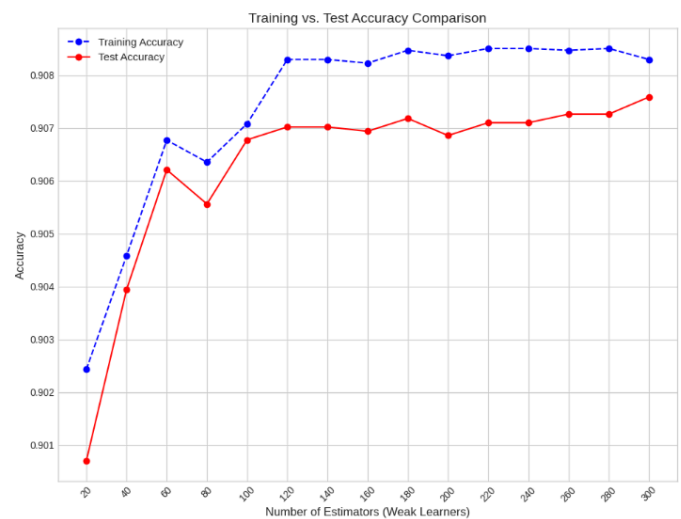
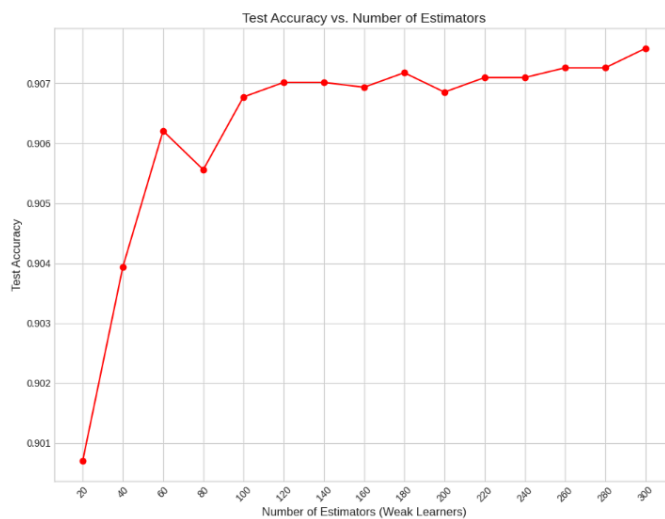


AdaBoost created a strong model from a sequence of weak learners (shallow trees). It iteratively corrects its mistakes, effectively driving down bias. AdaBoost can also overfit the training data, leading to an increase in variance. The experiments showed it achieved a strong accuracy of 0.9070. As with Random Forest, controlling the number of trees helps to improve model performance.

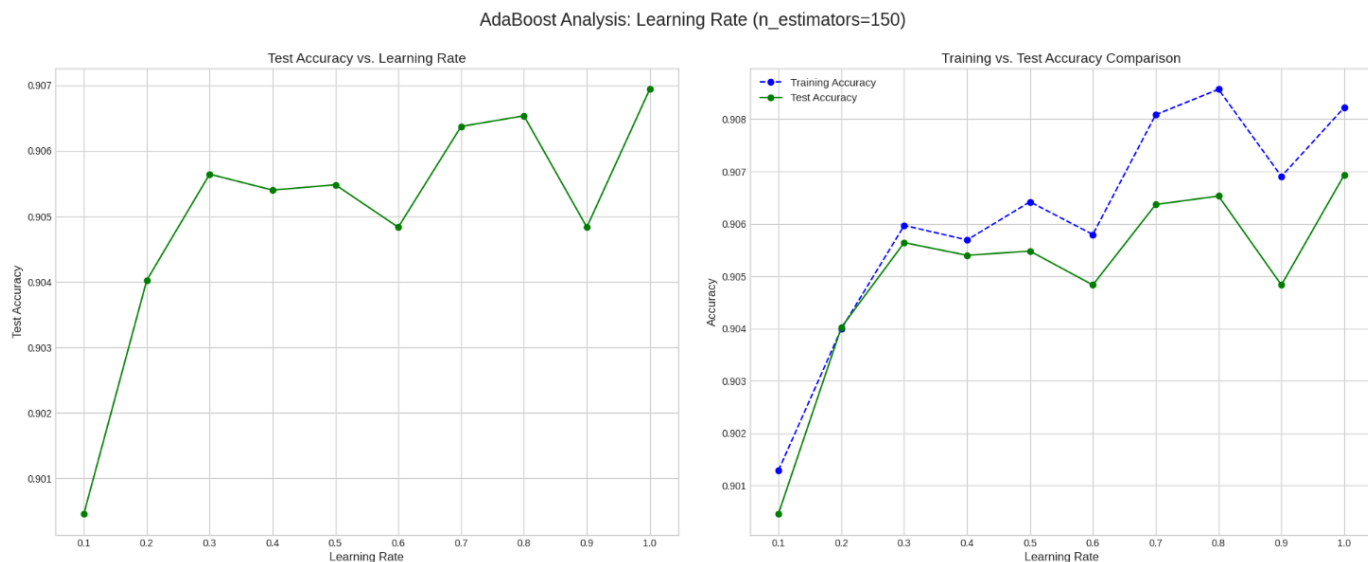
AdaBoost Results

The first experiment shows that increasing the number of estimators ($n_{\text{estimators}}$) effectively reduces initial bias, with performance rapidly improving before plateauing around 120 - 140 learners depending on where one wants to establish the cutoff for diminishing returns.

AdaBoost Analysis: Number of Weak Learners



The second experiment highlights the critical role of the `learning_rate` in controlling variance; a low rate (0.3) achieves a balance between training and test accuracies being almost identical. Higher learning rates produce a significant gap between the curves, indicating that while the model achieves high scores, it does so by overfitting the training data, making it less reliable and generalizable.



Optimal Model & Summary Results

Experiments are summarized in the table below:

Model Performance Summary					
		Hyperparameter Range Tested	Optimal Hyperparameter	Train Accuracy	Test Accuracy
Model	Experiment				
Decision Tree	Max Depth	max_depth in [None, 4]	max_depth = 4	0.9161	0.9118
	Min Leaf Size	min_samples_leaf from 1 to 50	min_samples_leaf = 35	0.9183	0.9140
Random Forest	N Estimators (Forest Size)	n_estimators from 1 to 200	n_estimators = 140	1.0000	0.9150
	Max Depth	max_depth in [None, 5 thru 30]	max_depth = 25	0.9850	0.9156
AdaBoost	N Estimators (Weak Learners)	n_estimators from 1 to 200	n_estimators = 120	0.9067	0.9070
	Learning Rate	learning_rate in [0.1, 0.3, 0.5, 0.8, 1.0]	learning_rate = 0.3	0.9069	0.9056

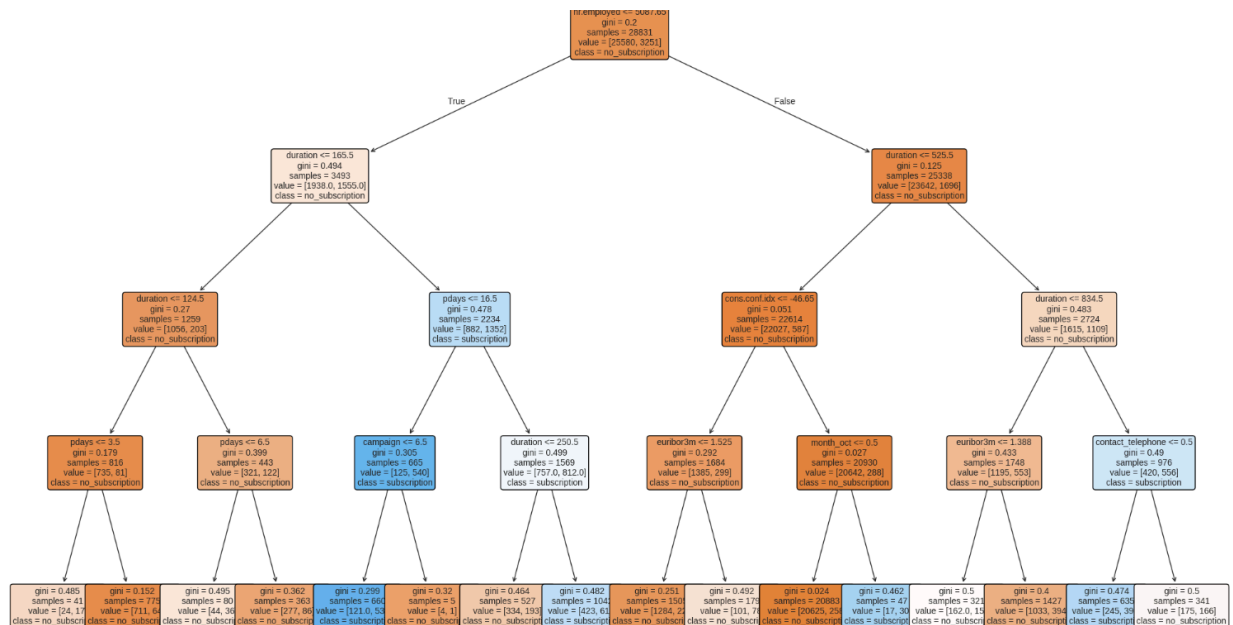
Based on these results, the optimal model is the Random Forest with n_estimators set to 100 with a max depth of 25, which achieved the highest test accuracy of 0.9156.

Recommendation:

For the primary goal of achieving the highest possible predictive accuracy, I recommend the deployment of the optimized Random Forest model. Its performance is attributed to the ensemble nature of the algorithm, effectively minimizing variance without sacrificing bias, leading to a generalizable model that should be monitored for any potential overfitting on new data.

It is important to acknowledge a significant limitation of the Random Forest model: it lacks interpretability. It is a *black box*, making it difficult to understand the underlying reasons for its predictions.

While the Random Forest is the choice for performance, the constrained Decision Tree (e.g., with `max_depth=4`) is significantly better for interpretability and explainability. This simpler model provides a clear, visualizable set of rules that can be easily communicated to stakeholders. It allows one to understand which factors are the most influential in predicting a client's subscription.



In summary, the best approach would be:

1. For Prediction: Use a Random Forest model constrained to 100 estimators at a max depth of 25 to maximize accuracy.
2. For Explanation: Use the constrained Decision Tree to understand and explain the key drivers behind the predictions.